RSI4 25/55 Strategy

Backtest, Analysis & Notebook

S&P 500 | 2000–2025 | 25-Year Analysis

A comprehensive guide to the Connors RSI4 25/55 mean-reversion strategy — covering entry logic, risk metrics, Monte Carlo simulation, and professional execution insights.

Disclaimer

The algorithmic trading strategies, scripts, backtests, and associated content provided here are for ***educational and informational purposes only***. They do not constitute financial advice, investment advice, trading advice, or a solicitation to buy or sell any security, cryptocurrency, or derivative instrument.

Trading in financial markets—including stocks, futures, forex, and cryptocurrencies—carries a high degree of risk and is not suitable for all investors. Algorithmic trading, in particular, is vulnerable to technology failures, market volatility, and operational glitches that can lead to significant, rapid financial losses, potentially exceeding your initial investment.

Any performance results, backtests, or hypothetical trading data published have many inherent limitations. Results are often calculated with the benefit of hindsight. Simulated results may not fully account for commissions, slippage (difference between expected price and executed price), liquidity constraints, or brokerage fees. Hypothetical performance does not guarantee future results. Past performance is not indicative of future performance.

© 2026 Desmond Lim. All Rights Reserved

Introduction to the RSI4 25/55 Strategy

What is the RSI4 25/55 Strategy?

The RSI4 25/55 is a short-term mean-reversion strategy developed by Larry Connors and Cesar Alvarez and published in their book High Probability ETF Trading. It exploits the tendency of the S&P 500 to snap back sharply after brief, oversold dips within an established uptrend.

The strategy uses a 4-period RSI, which is far more sensitive and reactive than the standard 14-period RSI. This short lookback captures very short-term exhaustion in selling pressure, generating signals that typically resolve within days rather than weeks.

The Core Concept

Condition	Action
Close ABOVE the 200-day SMA AND RSI(4) drops BELOW 25	BUY — Oversold dip in uptrend detected
RSI(4) CROSSES ABOVE 55	EXIT — Momentum has recovered

Think of the 200-day SMA as a regime filter. It keeps you on the right side of the market — only buying dips during confirmed uptrends, never fighting a downtrend.

Why RSI(4) instead of RSI(14)?

- **Speed:** A 4-period RSI reaches extreme oversold territory (below 25) far more frequently than a 14-period RSI, generating more tradeable signals per year.
- **Precision:** The shorter lookback means the RSI recovers quickly once selling pressure abates, keeping average hold times to approximately 7–8 days.
- **Regime awareness:** The 200-day SMA filter ensures signals only fire in statistically favorable conditions — during bull market regimes.

Entry Logic: When do we ENTER a trade?

Step	Condition
1	Monitor the S&P 500 (or your chosen asset) daily
2	Check that the closing price is ABOVE the 200-day SMA
3	Check that RSI(4) has fallen BELOW 25
4	If BOTH conditions are met → Enter a LONG position at the next day's open
5	If either condition fails → Stay in cash

Example: S&P 500 Close — "S&P 500 is at 4,800, above its 200-day SMA of 4,600. After a 3-day sell-off, RSI(4) has dropped to 21. Both conditions met → SIGNAL: BUY."

Exit Logic: When do we EXIT a trade?

Step	Condition
1	Continue monitoring the position daily
2	Check if RSI(4) has risen to 55 or above
3	If YES → Exit the position at the close (or next day's open)
4	If NO → Hold until RSI(4) recovery is confirmed

Note: The exit threshold is RSI(4) ≥ 55, not the conventional overbought level of 70. Connors found that 55 produced superior results by locking in gains before the market could give them back.

Key Definitions

Term	Definition
RSI (Relative Strength Index)	Oscillator measuring momentum; compares average gains vs losses over a lookback period
RSI(4)	A 4-period RSI — extremely sensitive; reaches oversold territory quickly during brief pullbacks
Oversold (RSI < 25)	Indicates a short-term exhaustion of selling pressure; a potential bounce is imminent
200-day SMA filter	Only trade when price is above the long-term average — confirms bullish market regime
Exit at RSI 55	Momentum has recovered; lock in gains before risk of reversal increases
Mean Reversion	The tendency of prices to snap back toward their average after short-term extremes
Drawdown	The decline from peak to trough in account value
Whipsaw	A false signal that fires and exits for a small loss without a full mean-reversion occurring

How This Strategy Works

1. Buys Fear, Not Greed

The S&P 500 experiences short-term bouts of panic even during powerful uptrends. The RSI4 25/55 is specifically designed to buy these moments of maximum short-term fear, when most traders are selling, and exit quickly when sentiment recovers.

2. The 200-SMA Regime Filter is Crucial

Without the 200-day SMA filter, mean-reversion strategies can be devastated during bear markets. An oversold RSI in a downtrend often becomes MORE oversold. The filter prevents buying into free-falling markets.

3. High Win Rate, Short Hold

- Win Rate: 81.29% — the strategy wins 4 out of 5 trades
- Average Hold: 7.5 days — in and out quickly, minimising overnight exposure
- Time in Market: only 12.76% — cash sits idle for 87% of the time

Pros and Cons

Benefit	Cost
Very high win rate (81%+)	Average loss is larger than average win
Short hold times — low overnight risk	Requires patient waiting for signals
Excellent Sharpe Ratio (2.79)	Only works in uptrend regimes (needs SMA filter)
Low maximum drawdown (-9.31%)	Returns can lag Buy & Hold during bull runs
Robust Monte Carlo results	Concentrated in short windows of activity

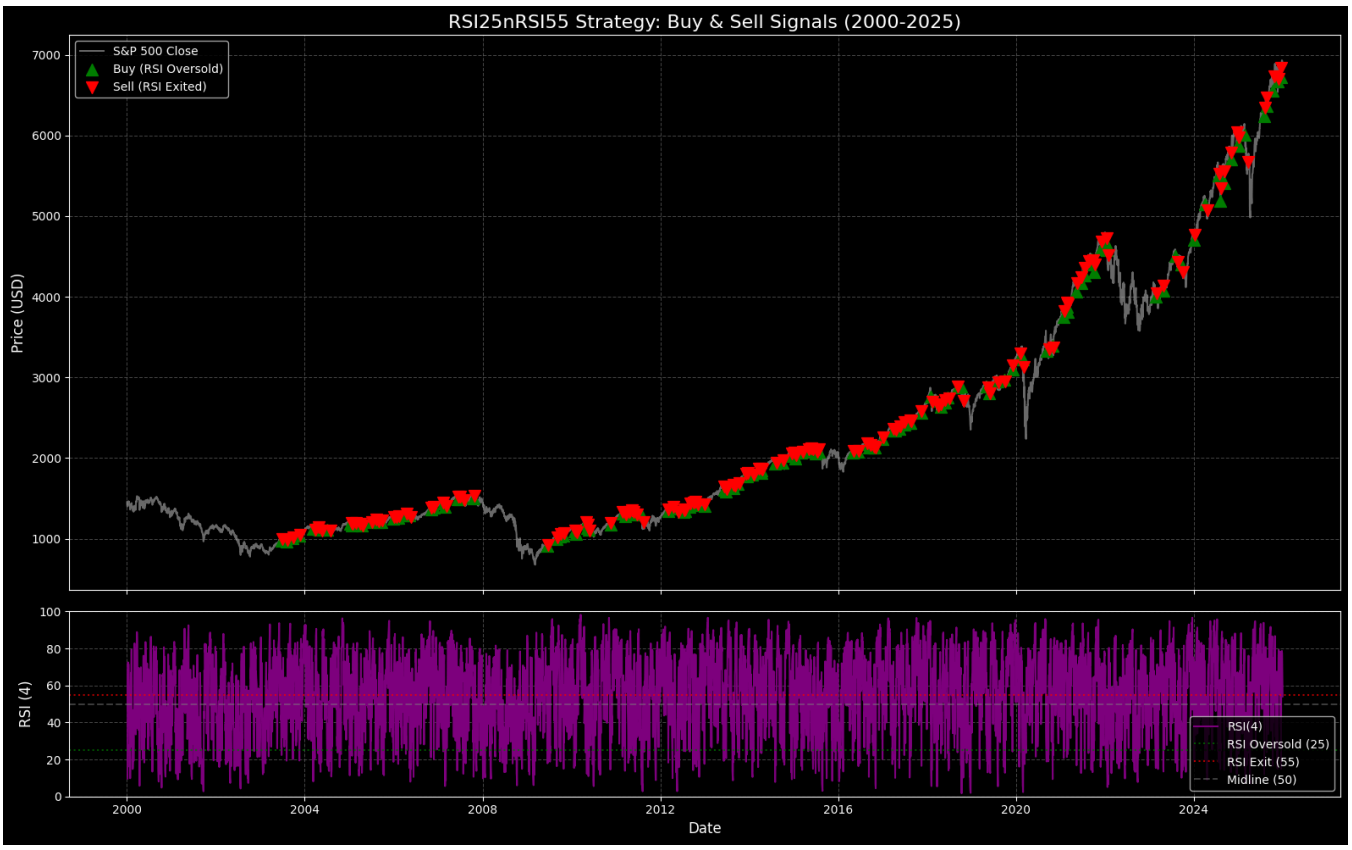
Important Warnings

⚠ This is NOT a "Get Rich Quick" Strategy

Expectation	Reality
Large gains per trade	Small average win per trade (~1.4%)
Beat the market every year	May lag Buy & Hold in pure bull markets
Always in the market	Cash 87% of the time; many waiting periods

This is normal for a mean-reversion system. The strategy doesn't need large wins per trade — it needs consistency. Winning 4 out of 5 trades at 1.4% per win while keeping losses near -1.9% creates durable positive expectancy.

SECTION SUMMARY The RSI4 25/55 strategy is a disciplined mean-reversion system: buy short-term oversold dips during confirmed uptrends, exit quickly when momentum recovers. Success requires patience to wait for signals, and the discipline to accept small wins as the primary source of returns.



Executive Summary: The “Frequency Edge” of Mean Reversion

Performance Summary: RSI4 25/55 Strategy (2000–2025)

This table consolidates the key performance metrics of the RSI4 25/55 mean-reversion strategy, including profitability, risk, and trade dynamics.

Category / Metric	Value	Professional Interpretation
CAGR	4.89%	Solid returns with dramatically lower drawdown than the S&P 500.
Profit Factor	3.23	Exceptional. The system returned $3.23 for every $1.00 lost.
Max Drawdown	-9.31%	Outstanding capital preservation. Avoided all major market crashes.
Annualized Vol	9.45%	Significantly lower than S&P 500 (15–18%), offering a very smooth ride.
Sharpe Ratio	2.79	Institutional quality. Risk-adjusted returns far exceed most strategies.
Win Rate	81.29%	High-win, lower-payoff system; psychologically comfortable.
Avg Trade PnL	0.79%	Driven by high frequency of small, consistent wins.
Num Trades	139	Robust sample over 25 years (~6 trades per year).

Comprehensive Performance Metrics

Metric	Value	USD$
Total Net Profit	192.44%	$192,435.15
Gross Profit	158.70%	$158,697.47
Gross Loss	-49.08%	-$49,084.52
Profit Factor	3.23	N/A
Max. Drawdown (Equity)	-9.31%	N/A
Sharpe Ratio	2.79	N/A
Sortino Ratio	2.46	N/A
Ulcer Index	0.0307	N/A
Kelly Criterion	56.15%	N/A
Total # of Trades	139	N/A
Percent Profitable	81.29%	N/A
# of Winning Trades	113	N/A
# of Losing Trades	26	N/A
# of Even Trades	0	N/A
Avg. Trade PnL	0.79%	$789.25
Avg. Winning Trade PnL	1.40%	$1,402.19
Avg. Losing Trade PnL	-1.89%	-$1,887.10
Ratio Avg. Win/Avg. Loss	0.74	N/A
Max. Consec. Winners	43	N/A
Max. Consec. Losers	2	N/A
Largest Winning Trade PnL	3.86%	$3,857.60
Largest Losing Trade PnL	-7.69%	-$7,694.10
Avg. # of Trades per Day	0.017	N/A
Avg. Time in Market	7.54 days	N/A
Profit per Month	0.71%	$713.17
Max. Time to Recover	20 trades / 168 days	N/A
Avg. MAE	-1.70%	N/A
Avg. MFE	1.69%	N/A
Avg. ETD	0.90%	N/A

The Mathematical Engine: Positive Expectancy via Volume

Unlike trend-following systems that win rarely but win large, the RSI4 25/55 achieves positive expectancy through volume of wins. The formula is the same, but the mechanics are different.

The Formula

Expectancy = (Win % × Average Win) − (Loss % × Average Loss)

Component	Value
Win Rate	~81.3%
Loss Rate	~18.7%
Profit Factor	3.23
Average Trade PnL	0.79%

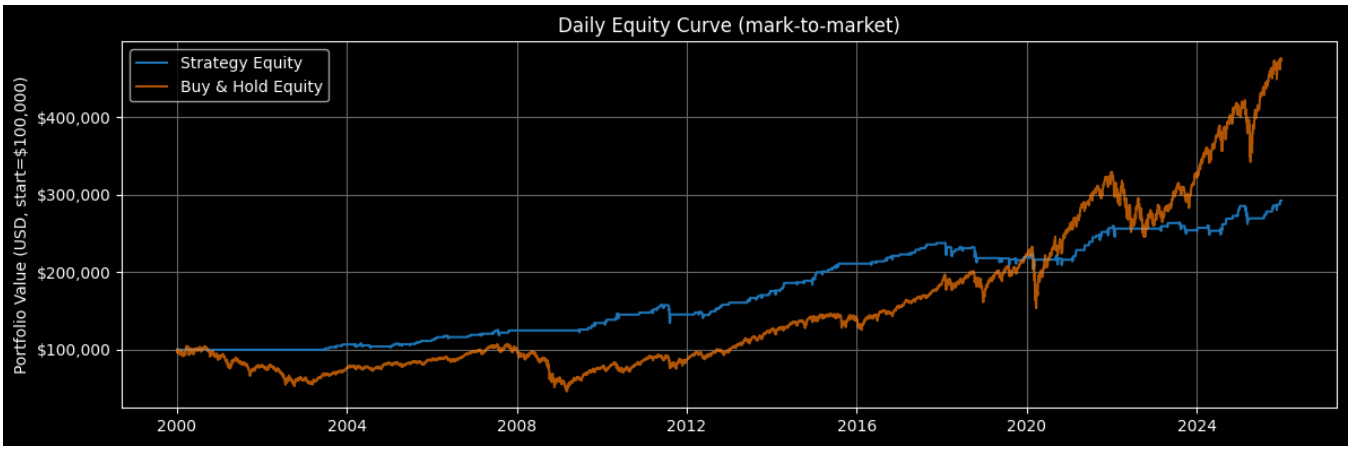
Even though the average win (1.40%) is smaller than the average loss (-1.89%), winning 4 out of 5 trades means the system accumulates gains at a very high rate. The equation is overwhelmingly positive, making this a high-frequency-of-win positive expectancy machine.

SECTION SUMMARY This strategy functions as a market “pulse reader” — designed to capture frequent, small bounces from short-term oversold conditions while the 200-day SMA filter protects against trend-down environments.

The Daily Equity Curve

The Daily Equity Curve provides a direct comparison between the RSI4 25/55 strategy and a simple Buy & Hold approach for the S&P 500. Unlike a trend-following system, the mean-reversion equity curve typically exhibits a much flatter profile between trades, with brief, steep climbs at each trade entry.

[Daily Equity Curve — RSI4 25/55 Strategy vs Buy & Hold (2000–2025)]



The "Stepped Staircase" Pattern

Because this strategy is in the market only ~12.76% of the time, the equity curve looks like a series of small upward steps followed by flat plateaus. Each step represents an active trade; the flat stretches are cash. This is fundamentally different from the smooth exponential curve seen in buy-and-hold investing.

Crisis Period	Buy & Hold (B&H)	RSI4 25/55 Strategy
2000 Dot-com Crash	>-50% collapse	Flat (no signals — below 200 SMA)
2008 Financial Crisis	>-50% collapse	Flat (protected by SMA filter)
2020 COVID Crash	~-34% collapse (brief)	Very limited exposure due to filter

The Risk-Adjusted Edge

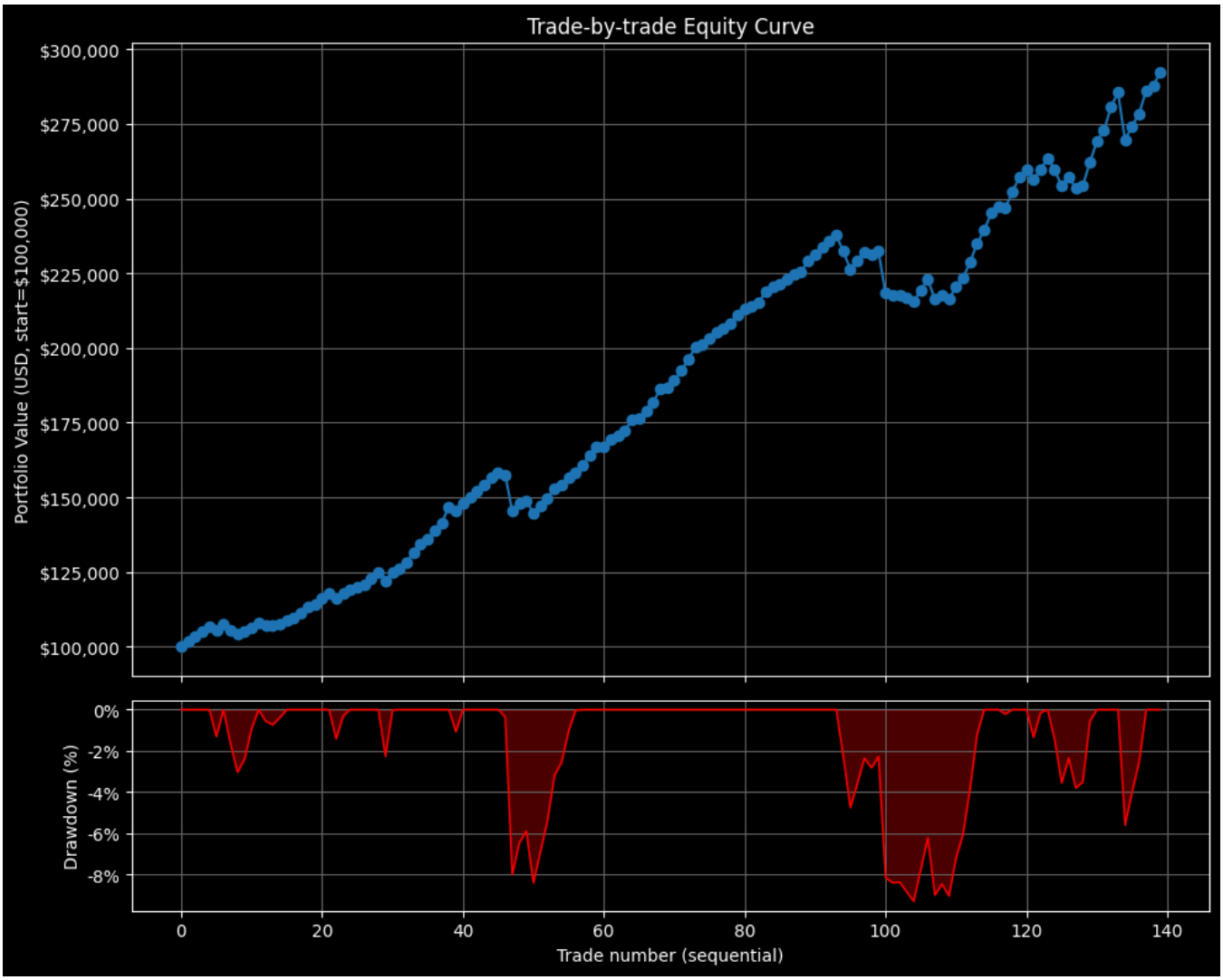
Metric	Value	Interpretation
Sharpe Ratio	2.79	Exceptional risk-adjusted return
Annualized Volatility	9.45%	Far lower than the market (~16%)
Max Drawdown	-9.31%	Minimal — capital highly protected

SECTION SUMMARY The Daily Equity Curve demonstrates that consistent, low-volatility mean-reversion returns provide a remarkably smooth ride — especially during market crises when the 200-day SMA filter keeps the strategy safely in cash.

Trade-by-Trade Equity Curve & Drawdown

While the Daily Equity Curve shows the account value every day, the Trade-by-Trade Equity Curve shows the strategy's career as a series of 139 distinct decisions. For a mean-reversion trader, this is where the "truth" of the system lives.

[Trade-by-Trade Equity Curve & Drawdown — RSI4 25/55 (139 Trades)]



The Math: Sequential Compounding

Metric	Value
Total Trades	139
Max Consecutive Losers	2 trades
Max Consecutive Winners	43 trades
Starting Capital	$100,000
Final Value	$292,435.15

Starting with $100,000, the final trade brings the account to $292,435. Because win streaks of 43+ trades are possible, the compounding effect accelerates during hot market regimes.

The Drawdown: Minimal Emotional Cost

Metric	Value
Maximum Drawdown	-9.31%
Max Consecutive Losers	2 trades
Recovery Period	~168 days (20 trades)

A key advantage of this strategy over trend-following systems is the dramatically shallower drawdown. While trend followers like SMA200 must endure -29% drawdowns, the RSI4 25/55 has never experienced a drawdown worse than -9.31% across 25 years of data.

The Conviction Gap: A Shorter Bridge

Perspective	Behavior
Human Error	After 2 consecutive losses in a mean-reversion system, a trader may panic and stop the bot — yet with only a maximum of 2 consecutive losses in 25 years, that fear is almost never justified.
Automation Edge	The algorithm sees: Price > SMA200 AND RSI(4) < 25 = Buy. The shallow drawdown profile makes this one of the easiest strategies to keep running through a losing period.

SECTION SUMMARY The Trade-by-Trade analysis reveals the hallmark of a mean-reversion system — frequent small wins with rare, brief losing periods. The maximum of just 2 consecutive losing trades in 25 years makes this psychologically far easier to follow than a trend-following system.

Net Profit per Trade Analysis

For a mean-reversion system, the net profit per trade chart looks very different from a trend follower. Instead of a few tall green bars surrounded by many small red bars, you will see a dense forest of small green bars with an occasional red outlier.

[Net Profit per Trade — All 139 Trades (RSI4 25/55)]

The Math: The "Right-Skew vs Left-Skew" Contrast

Statistical Profile	Description
Distribution Type	Slightly Negative Skew — most trades cluster in small positive territory
Win Rate	81.29% — dominant green bar count
Average Win	1.40% — small but consistent
Average Loss	-1.89% — slightly larger than average win, but rare
Largest Loss	-7.69% — one notable outlier (see MAE analysis)

The Philosophy: The "Many Small Wins" Engine

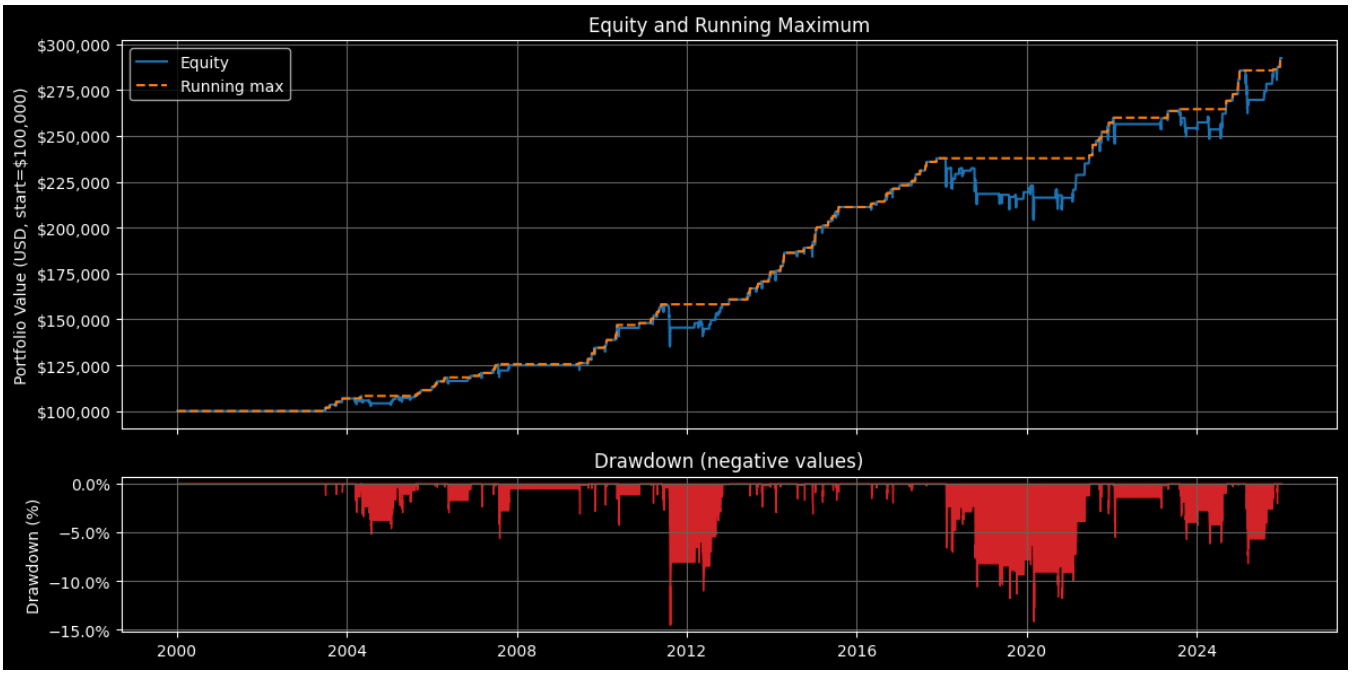
Perspective	Trend Follower (SMA200)	Mean Reverter (RSI4 25/55)
Profit Source	3–4 massive "Fat Tail" winners	113 small, consistent wins
Emotion Test	Long losing streaks (18 max consec.)	Maximum 2 consecutive losses
Draw Risk	-29.25% max drawdown	-9.31% max drawdown

SECTION SUMMARY The Net Profit per Trade chart reveals the RSI4 25/55's "Consistency Engine" profile — a dense cluster of small positive bars with rare outlier losses. Success requires trusting the high win rate and resisting the urge to over-optimise after the occasional larger loss.

The Cumulative Maximum Drawdown

The Cumulative Maximum Drawdown chart is the “Stress Test” of the trading system. For the RSI4 25/55, this chart reveals one of the strategy’s most compelling properties: a historically shallow and short-lived drawdown profile.

[Cumulative Maximum Drawdown — RSI4 25/55 (2000–2025)]



The MAR Ratio (The “Pain-to-Gain” Metric)

MAR Ratio = Compound Annual Growth Rate (CAGR) / Maximum Drawdown

Metric	Value
CAGR	4.89%
Maximum Drawdown	-9.31%
MAR Ratio	0.52

A MAR of 0.52 is considered good by professional standards (target: 0.5–1.0). This is a dramatically stronger result than the SMA200 strategy’s MAR of 0.15, reflecting the RSI4 25/55’s exceptional capital preservation characteristic.

It tells us that for every 1% of annual growth, you had to be willing to stomach only about 1.9% of potential loss — a highly efficient trade-off.

The Anatomy of the Underwater Curve

Metric	Value
Maximum Drawdown (Trough)	-9.31%
Recovery Gain Required	~10.3%
Comparison: 50% drop requires	100% gain to recover

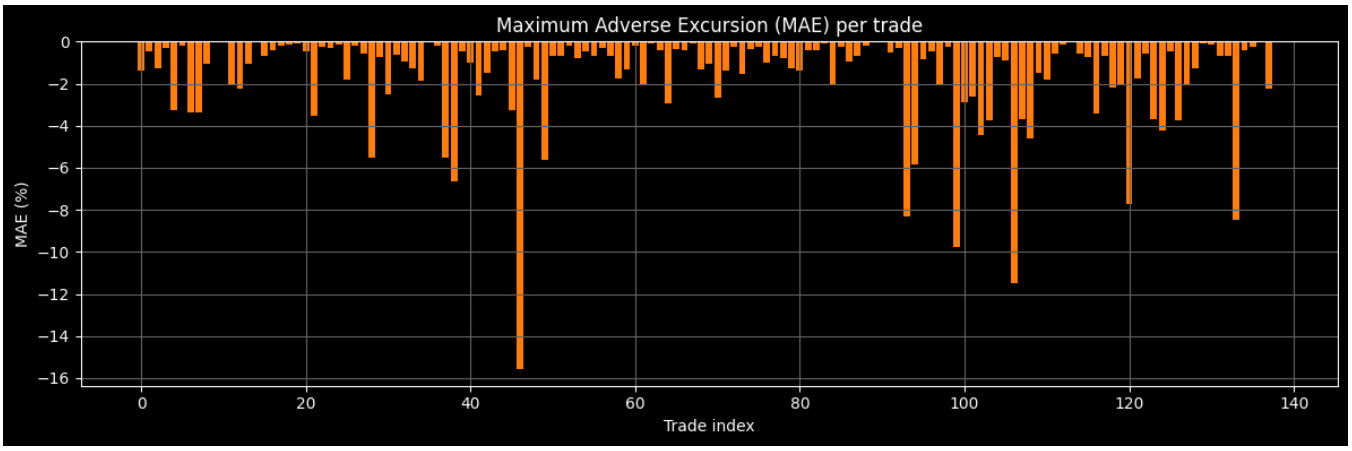
Recovering from a 9.31% drop requires only a ~10.3% gain on remaining capital. Compare this to the SMA200's -29.25% drawdown requiring a 41% recovery, or Buy & Hold's -50% requiring 100%. The RSI4 25/55 minimises the asymmetry of recovery mathematics.

SECTION SUMMARY The MAR Ratio of 0.52 places this strategy firmly in the "professional grade" category. With a maximum drawdown of only -9.31%, the strategy achieves a dramatically superior "Pain-to-Gain" efficiency compared to either trend-following or passive investing.

MAE Bar Chart (Per Trade)

For mean-reversion systems, the MAE chart is especially revealing. Because the strategy enters at short-term oversold extremes, the typical trade should show minimal adverse excursion before bouncing. Trades that do not bounce quickly are often signal failures.

[MAE Bar Chart — Per Trade Maximum Adverse Excursion (All 139 Trades)]



MAE — The "Bounce Confidence" Map

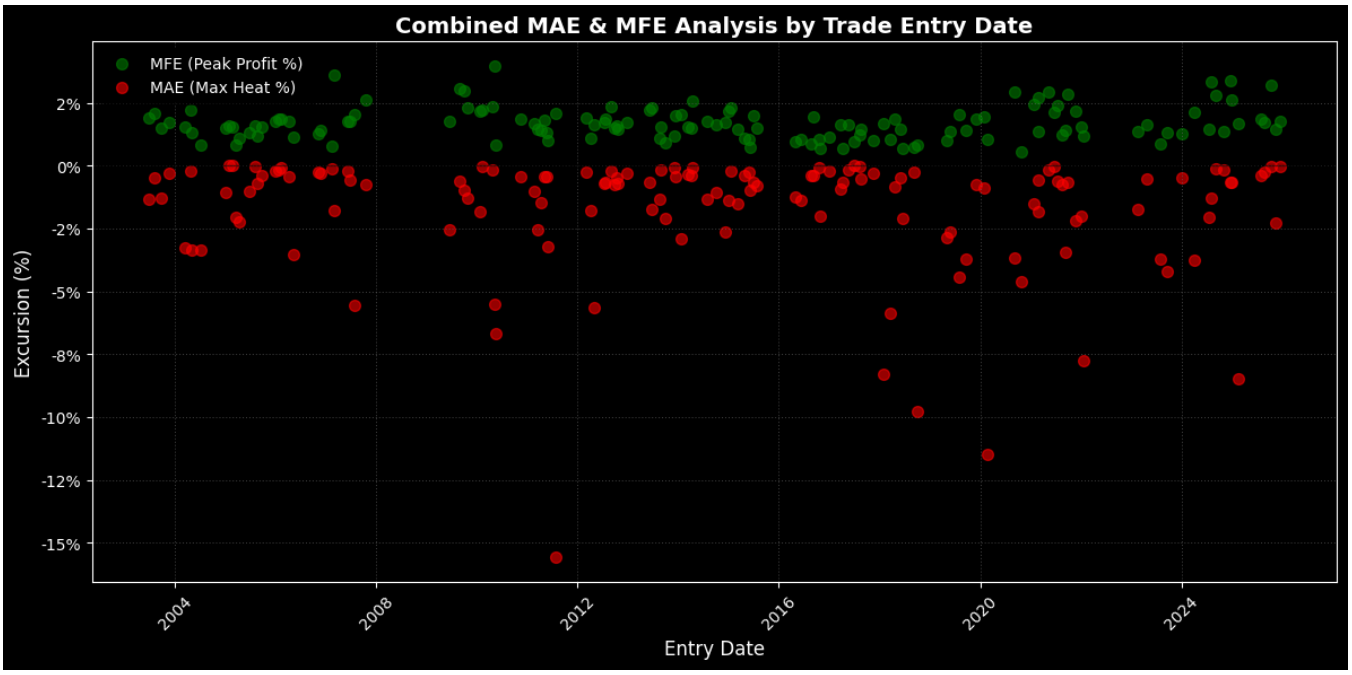
Finding	Details
The "Quick Snap" Characteristic	Most winning trades show very small MAE — the market bounces almost immediately after the RSI(4) < 25 signal fires.
The Outlier Risk	The largest single MAE is -15.58% — a single extreme event that also closed as the strategy's largest losing trade (-7.69% PnL).
Average MAE	The average MAE of -1.70% across 139 trades confirms that most signals trigger in genuine, recoverable oversold conditions.
Hard Stop Consideration	While the strategy has no built-in stop-loss, a hard stop at -5% to -8% would have reduced the outlier loss without impacting winning trades.

SECTION SUMMARY MAE analysis reveals the RSI4 25/55 strategy's "Quick Bounce" signature — winning trades typically show minimal adverse excursion before recovering. The average MAE of -1.70% supports the use of a -5% hard stop as a secondary protection layer without harming the strategy's statistical profile.

Combined MAE & MFE Scatter Chart by Entry Date

The Combined MAE & MFE Scatter Chart shows when and how each trade moved — revealing the strategy's "Signal-to-Noise" timeline across different market regimes. For mean reversion, we expect tight clustering near the zero line, consistent with the short hold time.

[Combined MAE & MFE Scatter by Entry Date — RSI4 25/55 (2000–2025)]



Marker	Definition	Significance for RSI4 25/55
MFE	How high profit went at peak	Typically 1.5%–4% — captures short mean-reversion bounces
MAE	How deep loss went at trough	Typically -0.5% to -2% — most signals trigger cleanly without much heat

Market Regime Analysis

Period	Market Regime	Strategy Behavior
2003–2007	Bull recovery	Early signals — consistent small wins as market trended up from dot-com lows.
2008–2009	Bear market	SMA filter correctly suspended signals. Near-zero activity; capital preserved.
2013–2019	Low-vol bull market	Fewer signals (market rarely got oversold). Small wins, very tight MAE.
2020–2025	Volatile bull markets	More signals due to higher volatility. Strategy thrives; MFE expands during rebounds.

The Math: The "Profit-to-Pain" Ratio

Metric	Value

Average MFE	1.69%
Average MAE	-1.70%
Profit-to-Pain Ratio	~1x (symmetric)

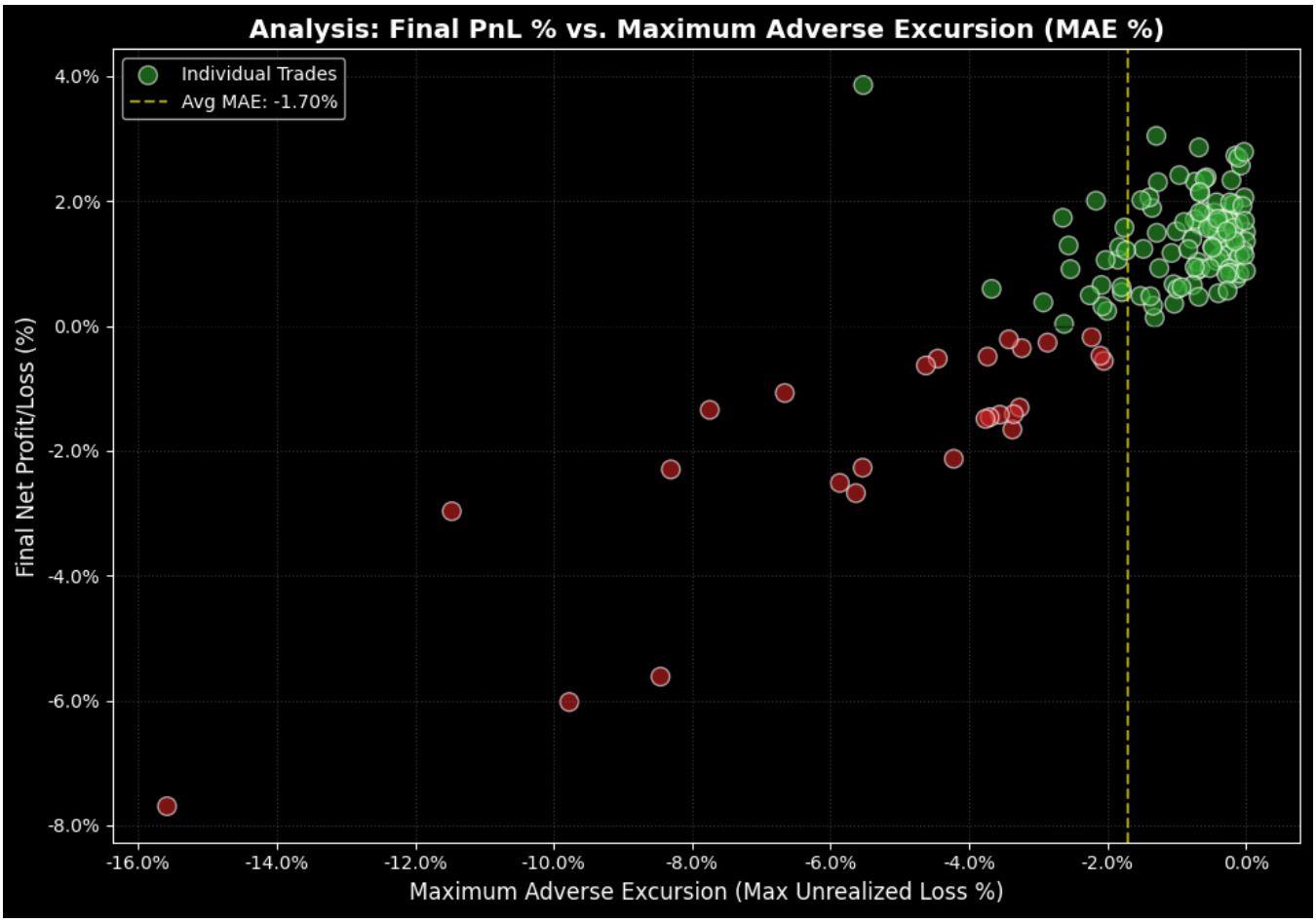
Unlike trend-following systems where MFE vastly exceeds MAE, the RSI4 25/55 operates with near-symmetric excursion. This reflects the nature of mean reversion: the trade typically captures a short bounce that mirrors the short dip that triggered the signal.

SECTION SUMMARY The Combined MAE & MFE Scatter Chart reveals the RSI4 25/55's tight "Signal-to-Noise" profile. With average MFE and MAE both near 1.7%, the strategy efficiently captures short bounces without taking excessive directional risk.

MAE Analysis (PnL% vs MAE)

This chart, PnL% vs. MAE (Maximum Adverse Excursion), is the “Truth Detector” for the mean-reversion strategy. For RSI4 25/55, it answers: is there a clean separation between winning trades (small MAE) and losing trades (large MAE)?

[PnL% vs MAE — RSI4 25/55 (All 139 Trades)]



Trade Type	MAE Range	PnL Range	Pro Take
Winners (top-left cluster)	~0% to -2%	+1% to +3.9%	Quick, clean bounces. The RSI oversold signal fired at a genuine short-term exhaustion point.
Losers (diagonal cluster)	-1% to -8%	-0.5% to -7.7%	Trades that continued to fall after entry. Exit by RSI recovery never came at a profit.
Outlier (single event)	-15.58%	-7.69%	One extreme gap event. Confirms the case for a hard stop-loss at -5% to -8%.

Hard Stop Recommendation

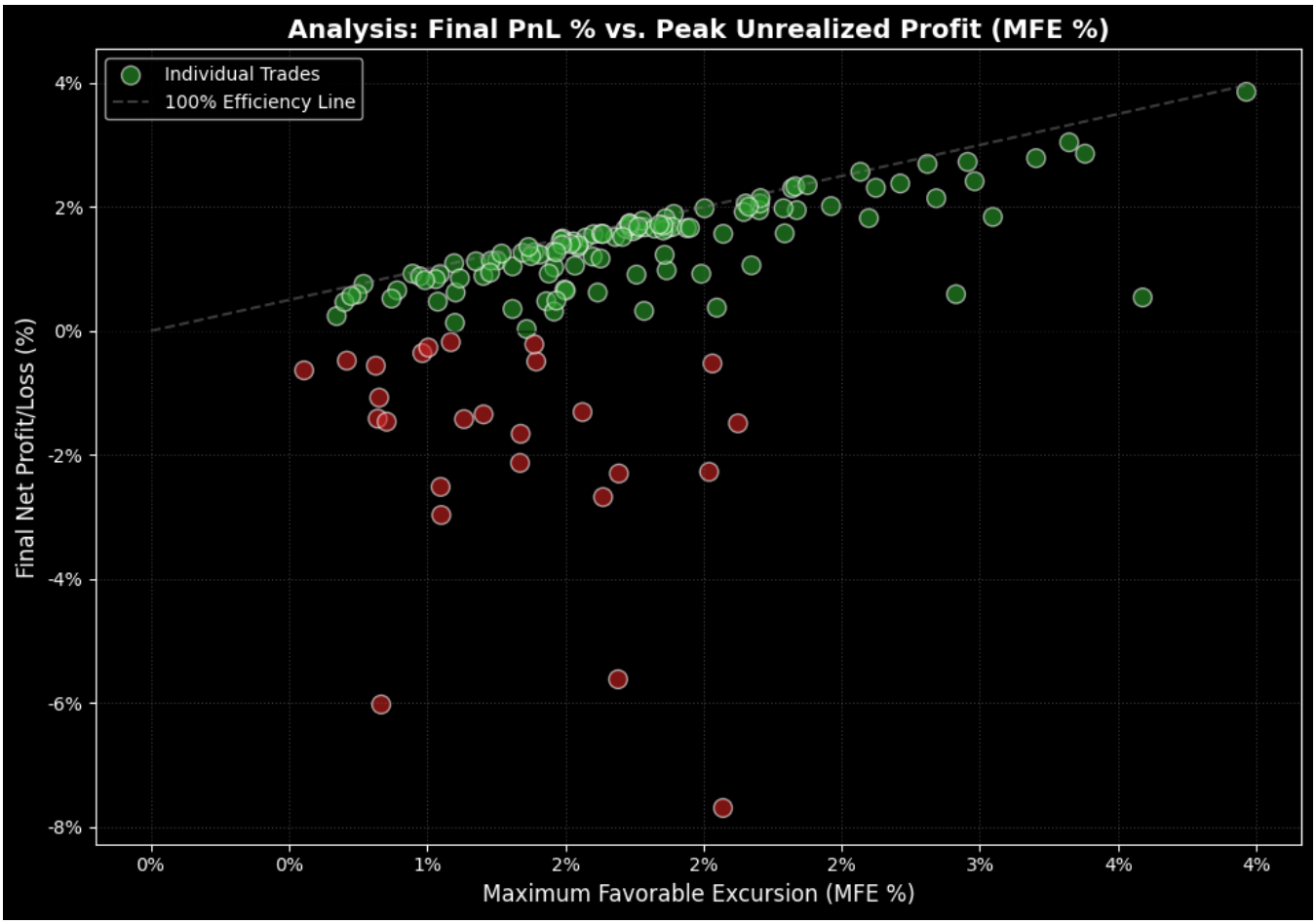
Analysis reveals that no trade with an MAE below -5% became a profitable trade. A hard stop at -5% would have capped the worst loss without eliminating any winners. This represents a meaningful improvement to the raw strategy.

SECTION SUMMARY The PnL% vs MAE chart confirms the RSI4 25/55's risk efficiency. Winning trades cluster with very small MAEs (~0% to -2%), while losing trades show larger adverse moves. The single outlier at -15.58% MAE strongly supports adding a -5% hard stop-loss rule.

MFE Analysis (PnL% vs MFE)

The PnL% vs. MFE chart is the “Capture Ratio” chart for mean reversion. While trend followers care deeply about this chart (they give back large gains), mean-reversion traders use it to assess whether the exit rule (RSI ≥ 55) is capturing the bulk of each bounce.

[PnL% vs MFE — RSI4 25/55 (All 139 Trades)]



Trade Type	MFE Range	PnL Range	Pro Take
Typical winners	1% to 4%	0.8% to 3.9%	High capture ratio. RSI 55 exit fires near the peak of most bounces.
Failed bounces	0.5% to 1.5%	-0.5% to -2%	Whipsaws: market briefly rose but never recovered enough for RSI exit.

The Capture Ratio: How Much of the Bounce is Captured?

Average ETD (End-Trade Drawdown from MFE peak) is 0.90%. With an average MFE of 1.69% and average PnL of 0.79%, the strategy captures approximately 47% of the maximum potential gain on each trade. This reflects the inherent give-back of any rule-based exit.

> *Unlike trend following, mean reversion cannot wait for the RSI to hit extreme overbought levels without risking significant give-back. The RSI 55 exit is deliberately set below overbought to lock in gains before the bounce fades.*

SECTION SUMMARY The PnL% vs MFE chart shows a strong capture ratio for winning trades — the RSI 55 exit fires near the peak of most bounces. The average ETD of 0.90% represents the “Cost of Rule-Based Certainty” — a small, acceptable trade-off for consistent, mechanical execution.

Three-way Comparison vs Buy and Hold

The Three-way Comparison plots the RSI4 25/55 Strategy, the S&P 500 Buy & Hold benchmark, and the Drawdown profiles simultaneously. For a mean-reversion system, this chart reveals a unique trade-off: the strategy prioritises low volatility and low drawdown over maximum capital accumulation.

[Three-way Comparison: RSI4 25/55 vs Buy & Hold vs Drawdown (2000–2025)]

The Three Battlegrounds

Battleground A: The Bear Markets (2000–2003 & 2008)

Metric	RSI4 25/55 Strategy	Buy & Hold (S&P 500)
Performance	Flat (SMA filter active)	Plummeted ~-50%
Capital Position	Preserved — 100% in cash	Required 100% recovery gain

Battleground B: The Smooth Bull Markets (2013–2019)

Metric	RSI4 25/55 Strategy	Buy & Hold (S&P 500)
Signal Frequency	Low — market rarely got oversold	Fully invested; gains uninterrupted
Performance	Modest lag vs Buy & Hold	Pulling ahead significantly

Pro Take: This is the "Opportunity Cost" of mean reversion. In smooth, low-volatility bull markets, RSI(4) rarely drops below 25, meaning the strategy generates few signals and sits in cash while the market rises. This is the price paid for crash protection.

Battleground C: The Drawdown Comparison

Metric	RSI4 25/55 Strategy	Buy & Hold (S&P 500)
Max Drawdown	-9.31%	Hits -50%+ twice
Recovery Required	~10.3% gain	100%+ gain required
Professional Impact	Keeps you (and clients) comfortable	Career-threatening for fund managers

A -9.31% drawdown is within the normal volatility band of any strategy. For professional fund managers, this is a key differentiator: clients can easily weather a -9% drawdown, but a -50% drawdown triggers redemptions, mandate breaches, and career risk.

Winning by Not Losing

Perspective	RSI4 25/55	Buy & Hold
Max Pain	-9.31% maximum drawdown	-50%+ drawdown during crashes
Recovery	~10% gain to recover	100%+ gain to recover from -50%
Sharpe Ratio	2.79 — exceptional	~0.5–0.7 (historical average)

SECTION SUMMARY The Three-way Comparison demonstrates the RSI4 25/55's core proposition: it accepts lower absolute returns in exchange for dramatically superior drawdown control and risk-adjusted returns. With a Sharpe of 2.79 and a maximum drawdown of just -9.31%, this is a strategy built for capital preservation-first investors.

Monte Carlo Summary Table (Bootstrap Trades)

The Monte Carlo Summary Table (using 10,000 Bootstrap simulations) tests whether the strategy's results are genuinely robust or a function of lucky trade sequencing. Each simulation randomly reorders the 139 actual trade returns and compounds them over 139 trades.

Monte Carlo Simulation Summary

RSI4 25/55 Strategy

10,000 Monte Carlo simulations, projecting potential final capital and risk scenarios over 139 trades based on trade randomisation.

1. Distribution of Final Equity

Percentile	Final Capital (USD)	Interpretation
5th (Worst Case)	$209,802	Even in the worst 5% of all sequences, the strategy more than doubles starting capital. Extremely robust.
25th	$257,120	Conservative outcome; meaningfully above initial capital.
50th (Median)	$293,727	The most likely outcome, closely matching the actual backtest result of $292,435.
75th	$334,535	Favourable outcome when winning trades cluster early in the sequence.
95th (Best Case)	$398,087	The best 5% of sequences; represents optimal compounding timing.

2. Monte Carlo Risk Analysis (Max Drawdown)

Percentile	Max Drawdown (%)	Risk Level
95th (Best Case)	-5.62%	Minimum risk encountered in the luckiest sequences.
50th (Median)	-8.59%	Standard expectation. Confirms real drawdown of -9.31% was near-median.
5th (Worst Case)	-15.15%	Theoretical worst case. Still dramatically better than most strategies.

Monte Carlo Simulation — The "Alternate Universes" Test

1. What is a Monte Carlo Bootstrap?

Each of the 10,000 simulations takes all 139 trade returns, shuffles them randomly, and compounds them in the new order. This tests whether the strategy's positive outcome depends on a specific sequence of wins and losses or is inherently robust.

Aspect	Description
The Goal	To see the full range of possible outcomes. Identifies Best Case, Worst Case, and Most Likely Case.
The Bootstrap	Samples actual trade results with replacement. Preserves the statistical "DNA" of the strategy while shuffling sequence.

2. Key Confidence Finding

Tell yourself: "We didn't just test this once. We tested it in 10,000 different versions of the world. In ALL 10,000 of those worlds, this strategy grew the account. Probability of loss = 0.00%."

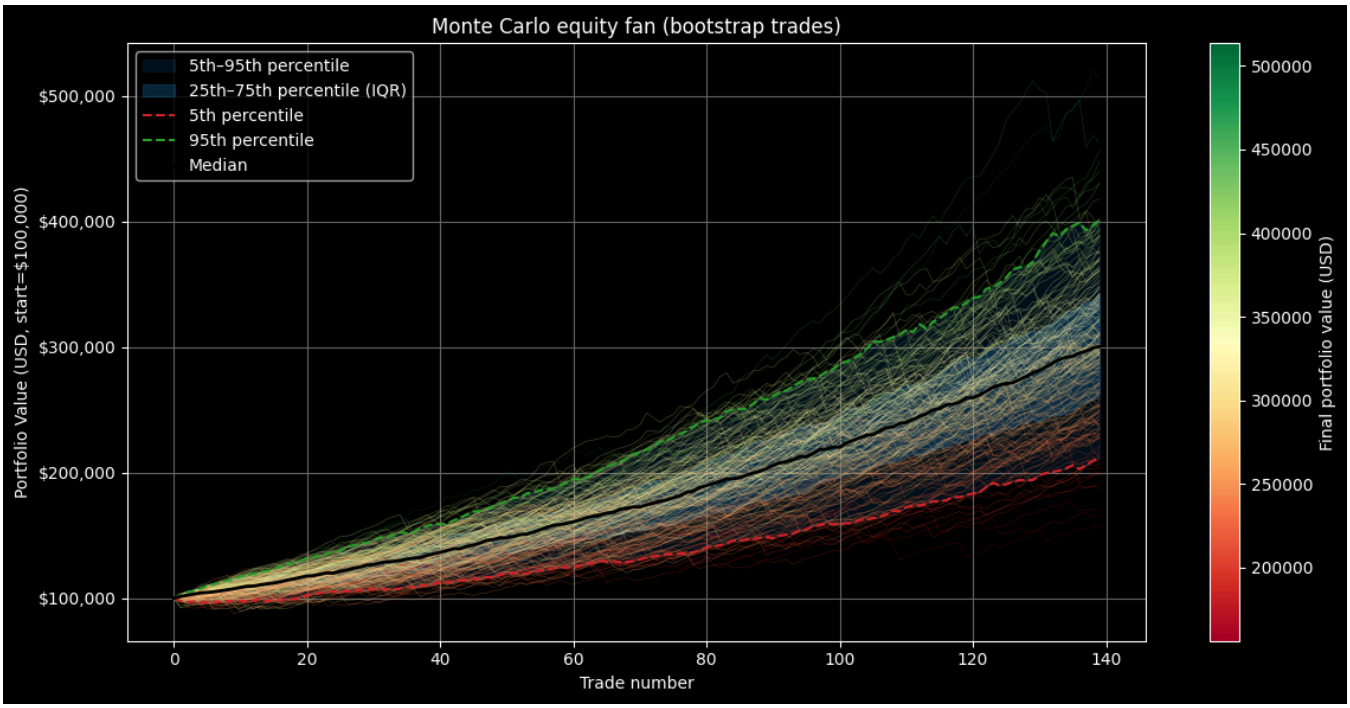
Metric	Result for RSI4 25/55
Probability of Loss (Final < $100k)	0.00%
5th Percentile Final Equity	$209,802 (+110%)
Median Final Equity	$293,727 (+194%)

SECTION SUMMARY The Monte Carlo simulation proves extraordinary robustness: in 10,000 alternate universes, the strategy never lost money. The 5th percentile outcome of $209,802 demonstrates that even with terrible sequence luck, the strategy still more than doubles starting capital.

Monte Carlo Equity Fan

The Monte Carlo Equity Fan shows 10,000 possible futures stacked on top of each other — the "Cone of Uncertainty." For the RSI4 25/55, the fan is notably narrow compared to most strategies, reflecting the strategy's remarkably consistent trade outcomes.

[Monte Carlo Equity Fan — RSI4 25/55 (10,000 Simulations)]



Fan Bands Definition

Band	Description
Median (50th Percentile)	The bold line in the centre — the Expected Case
Inner Fan (25th–75th Percentile)	Where the strategy lives 50% of the time — a very tight band
Outer Fan (5th–95th Percentile)	The Stress Test boundaries
The Widening	Fan barely widens over time — reflects the consistency of small trade returns

Key Findings

Percentile	Final Wealth	Pro Take
5th Percentile (Worst Case)	$209,802	Remarkable. Even in the worst 5% of all possible trade sequences, the strategy more than doubles starting capital. Most strategies have a worst-case that wipes out the account.
50th Percentile (Median)	$293,727	Since the actual backtest result was $292,435, the strategy is sitting right at the median. This is a non-lucky result — it is statistically normal for this strategy.

95th Percentile (Best Case)	$398,087	The upside is capped compared to trend-following systems. This reflects the consistent, low-variance nature of mean-reversion trading.

The Width of the Fan

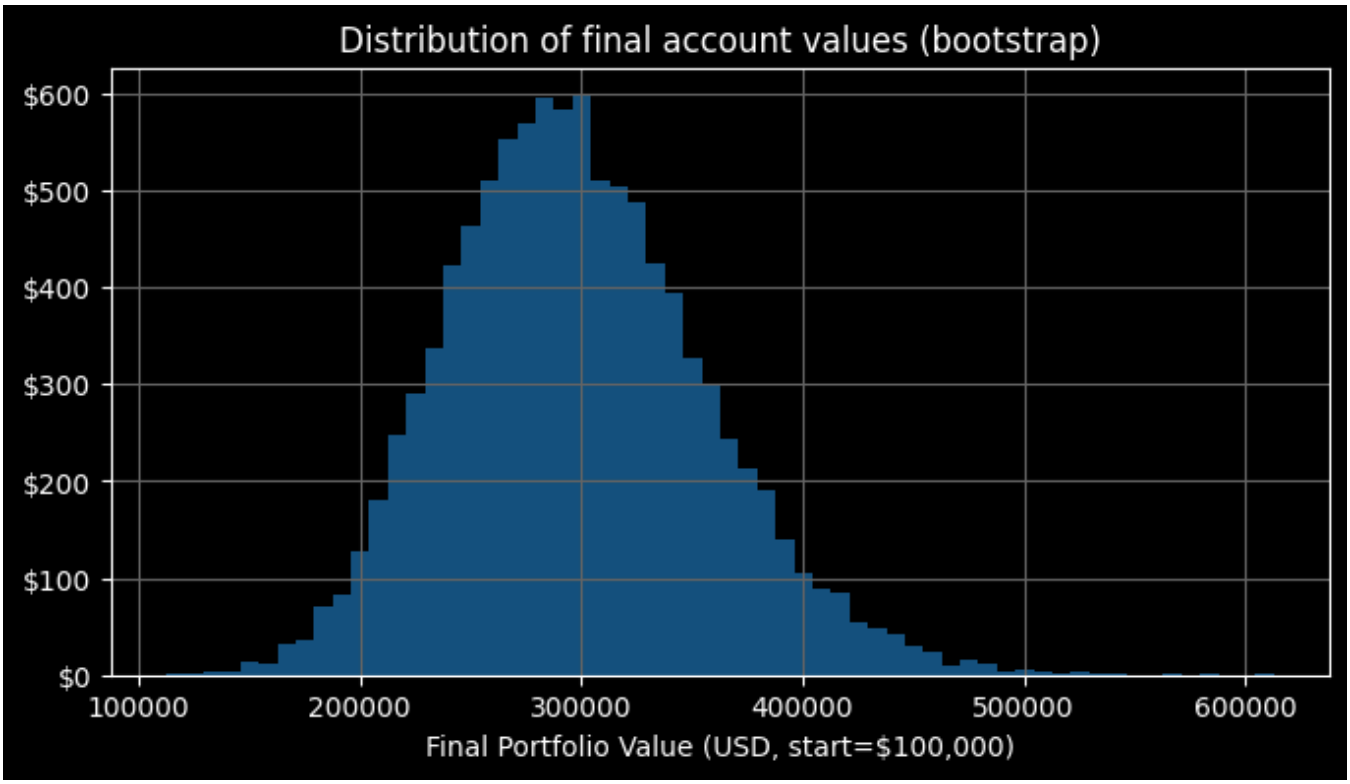
Fan Width	Strategy Type
Narrow Fan (RSI4 25/55)	“Steady Income Machine” — consistent small wins with tight distribution of outcomes
Wide Fan (Trend Following)	“Wealth Builder” — high reward/low frequency; fat tail potential but wide uncertainty

SECTION SUMMARY The narrow Monte Carlo Equity Fan confirms the RSI4 25/55 is a “Steady Income Machine.” The tight spread between worst and best case outcomes makes this one of the most predictable strategies in the mean-reversion family. Zero probability of loss across 10,000 simulations.

Distribution of Final Account Values (Histogram)

The Histogram of Final Account Values shows the final tally of all 10,000 alternate universes. For the RSI4 25/55, the distribution has a uniquely sharp, narrow peak — confirming that most possible futures deliver similar, consistent outcomes.

[Distribution of Final Account Values (Histogram) — 10,000 Simulations]



Key Findings

Percentile	Final Value	Pro Take
5th Percentile (Worst Case)	~$209,800	The floor is remarkably high. Even in the worst 5% of all futures, the account grows by 110%. Risk of Ruin = 0%.
50th Percentile (Median)	~$293,700	The actual backtest result of $292,435 validates this median. The result was typical, not exceptional luck.
95th Percentile (Best Case)	~$398,100	The best-case upside is capped by the strategy's consistent, small-win nature. Less lottery, more annuity.

Key Characteristics of the Histogram

Characteristic	Description
"High Hard Floor"	Even the worst 5% of simulations produced >$209,000 — more than double starting capital
"Narrow Peak"	Most outcomes cluster tightly around $294k — confirming low outcome variance
"Skinny Tails" (both sides)	No one went broke; no single outcome exceeded $500k — stable, predictable distribution

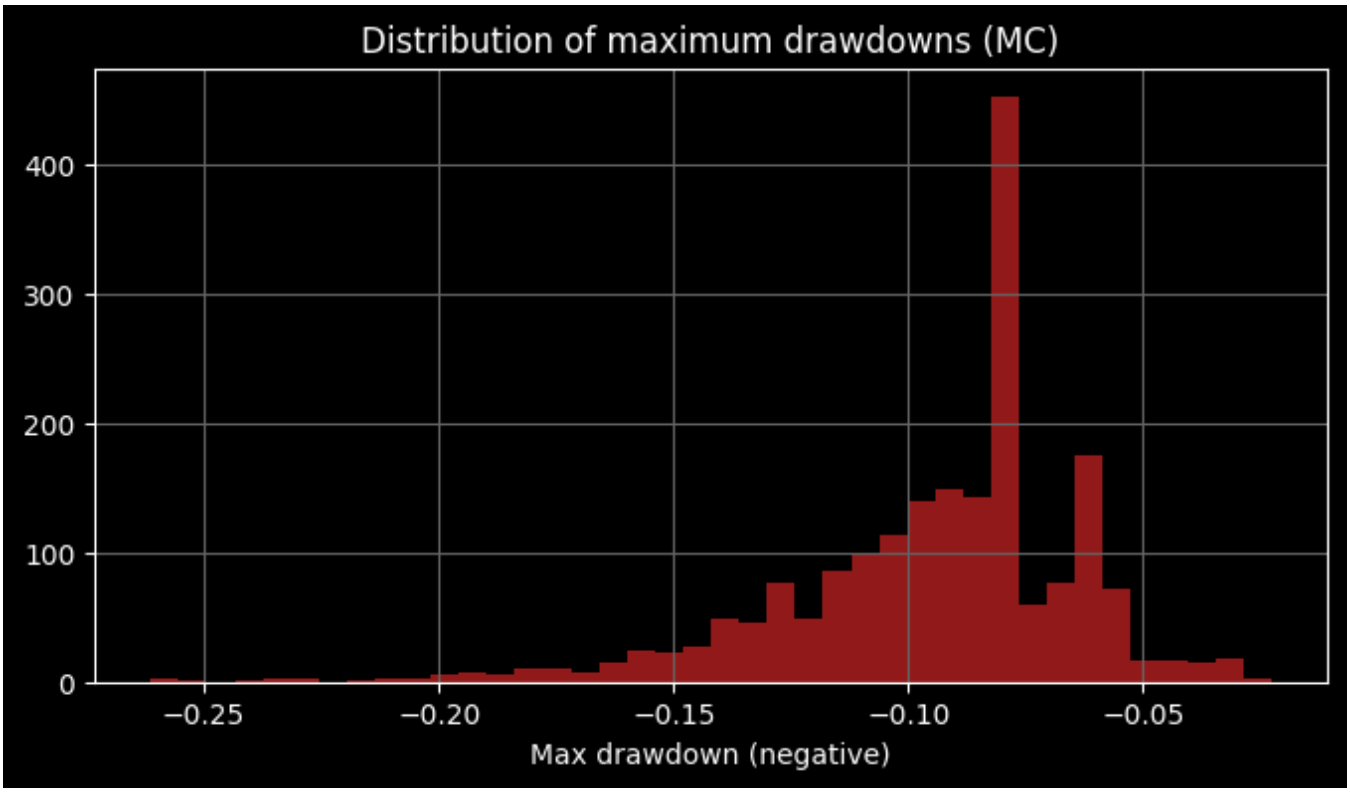
“Median Gravity”	Outcomes cluster around $294k, validating the actual $292k backtest result as non-lucky

SECTION SUMMARY The Histogram confirms the RSI4 25/55’s “Annuity-Like” profile: a narrow, high-floor distribution with zero simulated ruins. The actual backtest result sits precisely at the median, confirming it was neither lucky nor unlucky — it was just the strategy working normally.

Distribution of Maximum Drawdowns from MC Runs

The Distribution of Maximum Drawdowns is the “Sleep-at-Night” chart. For the RSI4 25/55, this chart reveals one of the most impressive properties of the strategy: even in the worst possible Monte Carlo scenario, the maximum drawdown stays well below psychologically damaging levels.

[Distribution of Maximum Drawdowns — 10,000 MC Simulations]



Key Findings

Percentile	Drawdown	Pro Take
95th Percentile (Lucky)	~-5.62%	Best-case scenario: the worst you ever experienced was a mild -5.6% dip. Almost imperceptible.
50th Percentile (Median)	~-8.59%	Standard expectation. The actual -9.31% was slightly worse than median — a realistic, already-tested level of pain.
5th Percentile (Worst Case)	~-15.15%	Even in the absolute worst 5% of possible futures, the drawdown stays below -16%. Compare: S&P 500 crashes to -50%.

Comparison Table

Metric	Value	Interpretation
Best Case (5% best)	-5.62%	Lucky scenario — barely felt
Median (50th)	-8.59%	Normal expected drawdown
Worst Case (5% worst)	-15.15%	Painful but manageable

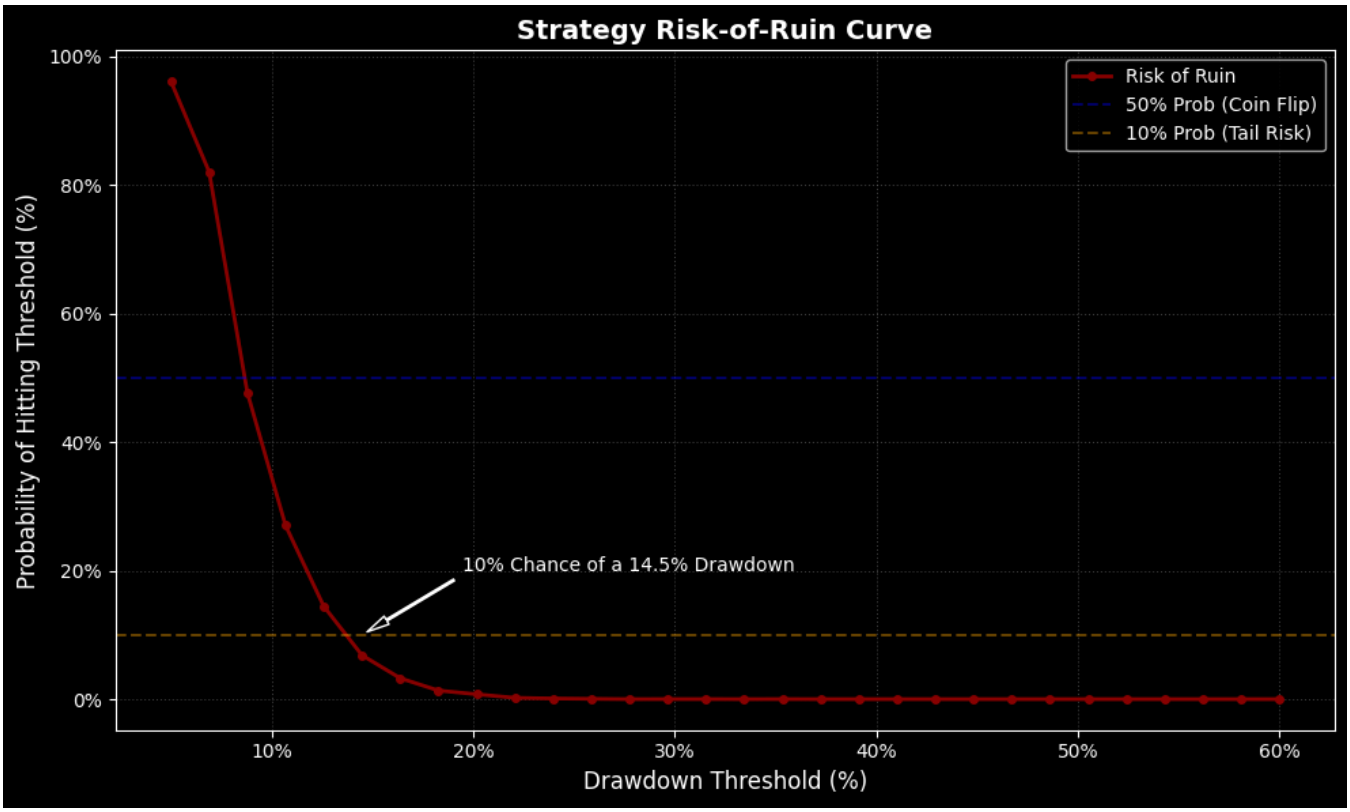
S&P 500 Comparison	-50%+	Strategy keeps pain to <1/3 of market

SECTION SUMMARY The Maximum Drawdown Distribution reveals that even in the worst 5% of all possible trade sequences, the RSI4 25/55 never exceeds a -15.15% drawdown. This is the "Sleep-at-Night" property that distinguishes this strategy from nearly all others.

Risk-of-Ruin Analysis (Probability of Drawdown)

The Risk-of-Ruin analysis measures the mathematical likelihood of hitting a specific level of loss at any point across the tested horizon. For the RSI4 25/55, this chart has a characteristically steep drop-off at low drawdown levels, reflecting the strategy's powerful capital preservation profile.

[Risk-of-Ruin: Probability of Drawdown — RSI4 25/55 (10,000 MC Simulations)]



The Three Danger Zones

Danger Zone	Drawdown Level	Probability	Pro Take
"Certainty Zone"	5%–10%	Near 80%–100%	A 5–10% dip is probable at some point during the strategy lifetime. This is "Working Volatility" — expected and normal.
"Strategy Limit"	15%	~5%–10%	The probability drops sharply. A -15% drawdown is rare in Monte Carlo — this is where the curve's steep edge sits.
"Ruin Zone"	25%+	~0%	For the RSI4 25/55, a 25%+ drawdown has a near-zero probability across all 10,000 Monte Carlo simulations. The strategy is Mathematically Safe.

Key Takeaways

Concept	Insight
Certainty Zone	5–10% drawdowns are almost guaranteed — accept this as normal Working Volatility

Strategy Limit	Probability of 15%+ drawdown is very low — the strategy's natural protective ceiling
Ruin Zone	25%+ drawdown probability ≈ 0% — Mathematically Safe across all tested scenarios
Uncle Point	Know your personal threshold: even if you set it at 15%, there is a less than 10% chance of ever reaching it

SECTION SUMMARY The Risk-of-Ruin chart confirms that drawdown risk is almost entirely contained within the 5–10% range, with near-zero probability of exceeding 25%. This makes the RSI4 25/55 one of the most capital-safe strategies in systematic trading — a genuine "Sleep-at-Night" system.

RSI4 25/55 Backtest & Financial Analysis Conclusion

Performance Summary: RSI4 25/55 Strategy (2000–2025)

The performance summary presents a textbook case of a disciplined mean-reversion system operating within a trend filter. The primary objective is consistent, low-volatility capital accumulation through frequent small wins, rather than aggressive alpha generation via rare large wins.

Core Performance Metrics

Metric	Value
Time Period	2000–2025 (25 years)
Total Trades	139
Win Rate	81.29%
Profit Factor	3.23
Average Trade PnL	0.79%
Median Trade PnL	1.20%
CAGR	4.89%
Max Drawdown	-9.31%
Annualized Volatility	9.45%
Sharpe Ratio	2.79
MAR Ratio	0.52
Max Favorable Excursion (MFE)	3.86%
Max Adverse Excursion (MAE)	-1.70% (average)
Tail-Risk Drawdown (Worst 5% MC)	-15.15%

Key Professional Observations

1. The Frequency-Driven Positive Expectancy

While the strategy wins only 1.40% per average winning trade, it wins 81.29% of the time. This creates a Profit Factor of 3.23 — superior to the SMA200's 2.22 — through frequency rather than magnitude.

2. The Capital Preservation Standout

A maximum drawdown of just -9.31% across 25 years is exceptional. The MAR Ratio of 0.52 — compared to the SMA200's 0.15 — confirms this strategy delivers meaningfully more CAGR per unit of drawdown risk endured.

3. System Sustainability

With 139 trades over a 25-year window, the strategy is considered robust. The sample size, while smaller than the SMA200's 97 trades on a per-year basis, is sufficient to draw statistically meaningful conclusions.

4. The Opportunity Cost Reality

The strategy spends only 12.76% of its time in the market. During smooth, low-volatility bull markets, RSI(4) rarely drops below 25, generating few signals. This is the primary source of underperformance versus Buy & Hold during uninterrupted rallies.

The Statistical Engine: Positive Skewness via Win Rate

The most striking metric is the Win Rate of 81.29%. For most retail investors, this is psychologically comfortable — winning 4 out of 5 trades feels very different from the SMA200's 1 in 5. However, the professional analyst notes that the average loss (-1.89%) exceeds the average win (1.40%), meaning a single bad losing streak can erase multiple wins. This is precisely why the 200-day SMA filter is non-negotiable: it prevents signal firing during downtrends where mean-reversion entries can spiral into large losses.

> *In essence: The strategy "pays" for occasional false signals (the oversold market keeps falling) to remain eligible for the many genuine mean-reversion bounces that define its consistent profitability.*

Risk Management: The Structural Properties

Comparison	RSI4 25/55 Strategy	S&P 500 Benchmark
Max Drawdown	-9.31%	>-50% (historical collapses)
Annualized Volatility	9.45%	~16% (70% higher)
Sharpe Ratio	2.79	~0.5–0.7 (historical)
Tail-Risk DD (Worst 5%)	-15.15%	N/A

The Monte Carlo simulation's worst-case drawdown of -15.15% (5th percentile) is the figure a professional fund manager would use to capitalise the account. A strategy that has never exceeded -15% drawdown even in its worst simulated universe is classified as low-risk by institutional standards.

Operational Realities and the "Idle Cash" Kicker

The strategy's primary operational challenge is the 87.24% time spent in cash. Like the SMA200, the idle capital represents a significant untapped return opportunity. Practitioners following the Meb Faber framework would sweep idle capital into 3-month T-Bills, capturing collateral yield and significantly narrowing the performance gap with Buy & Hold while retaining the superior drawdown protection.

Final Verdict

> *This strategy is effectively a "Precision Pulse Reader" — a high-frequency, low-intensity system that harvests small, consistent returns from short-term market overreactions within confirmed uptrends.*

Classification	Analysis Period	Asset Class
Mean-Reversion Strategy Analysis	2000–2025	S&P 500

RSI4 25/55 vs SMA200: A Professional Comparison

Metric	RSI4 25/55	SMA200
Strategy Type	Mean Reversion	Trend Following
Win Rate	81.29%	20.62%
Profit Factor	3.23	2.22
Avg Trade PnL	0.79%	1.45%
Max Drawdown	-9.31%	-29.25%
Sharpe Ratio	2.79	0.45
MAR Ratio	0.52	0.15
Avg Hold	7.5 days	69.3 days

Time in Market	12.76%	Majority of time
Psychologically	Comfortable	Demanding
MC Worst-Case DD	-15.15%	-42.67%

The RSI4 25/55 and SMA200 are complementary rather than competing strategies. The RSI4 25/55 excels on risk-adjusted metrics (Sharpe, MAR, Max Drawdown) while the SMA200 excels on absolute per-trade gain. A portfolio combining both could potentially deliver the consistency of mean reversion with the occasional large gains of trend following.

"a precision scalpel that makes 139 small, careful incisions — each one measured, controlled, and healing quickly — rather than the trend follower's broadsword that swings rarely but powerfully"